

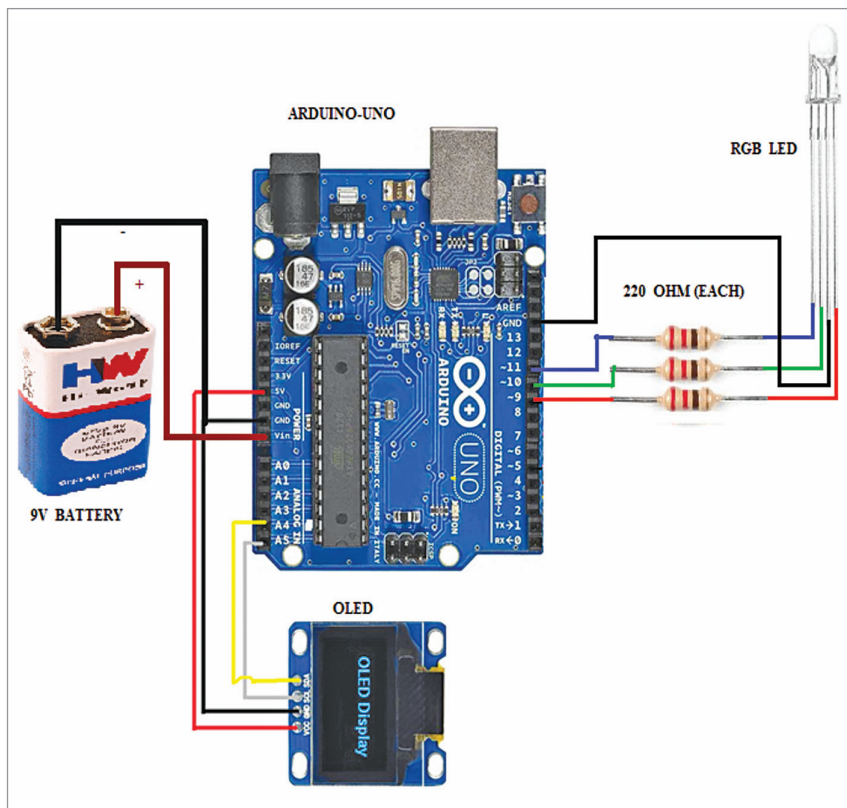


Fig. 2: Circuit diagram

```

colorbuddy.ino
1 //RAKESH JAIN PROGRAM FOR "Colour Buddy: Learn Colours with Lights"
2 #include <Adafruit_GFX.h>
3 #include <Adafruit_SSD1306.h>
4 #include <Wire.h>
5 // OLED settings
6 #define SCREEN_WIDTH 128
7 #define SCREEN_HEIGHT 64
8 #define OLED_RESET -1
9 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
10 // RGB LED pins
11 int redPin = 9;
12 int greenPin = 10;
13 int bluePin = 11;
14 // Define structure for color
15 struct Color {
16     byte r, g, b;
17     const char* name;
18 };
19 // Updated and more visible color list
20 Color colors[] = {
21     {255, 0, 0, "Red"},
22     {0, 255, 0, "Green"},
23     {0, 0, 255, "Blue"},
24     {255, 255, 0, "Yellow"},
25     {255, 165, 0, "Orange"},
26     {255, 20, 147, "Pink"},
27     {0, 255, 255, "Cyan"},
28     {255, 255, 255, "White"},
29     {0, 0, 0, "Black"}, // Will not light up LED means showing
30     {128, 0, 128, "Purple"},
31     {128, 0, 0, "Maroon"}

```

Fig. 3: Snippet of the source code

TABLE 2
COLOUR COMBINATIONS
OF RGB LED

Colour Name	Red	Green	Blue
Black	0	0	0
Red	255	0	0
Green	0	255	0
Blue	0	0	255
Yellow	255	255	0
Cyan	0	255	255
Magenta	255	0	255
White	255	255	255
Orange	255	165	0
Pink	255	192	203
Purple	128	0	128
Brown	165	42	42
Gray	128	128	128
Light Blue	173	216	230
Lime	0	255	0
Dark Green	0	100	0
Maroon	128	0	0
Navy	0	0	128
Olive	128	128	0
Teal	0	128	128

colour-based feedback. Together, this arrangement can be applied to interactive or decorative systems where both visual display and coloured light indications are required.

Software

The software for this system is developed in the Arduino IDE, using C/C++ programming to control the RGB LED and display information on the OLED screen. The RGB LED is driven using three PWM pins (9, 10, and 11), allowing brightness control of each colour channel to produce a wide range of colours. The code includes functions to set individual LED colours and create transitions or effects. Timing functions like delay() can be used to control the speed of colour changes, while loops handle repeated sequences.

For the OLED display, the Adafruit SSD1306 and Adafruit GFX libraries are used to initialise and draw text, shapes, or images.